

Master of Science
in

Knowledge and Technology Management



THE HONG KONG
POLYTECHNIC UNIVERSITY

香港理工大學

Department of Industrial
& Systems Engineering

Managing Knowledge (ISE 542)

Lecture 1: The Evolution of Knowledge



Learning Outcome

- To learn the evolution of the concept of knowledge
- To understand and differentiate various types of knowledge
- To explain the way in which knowledge can be converted (Nonaka's cycle of knowledge conversion)
- To learn the concept of metaphors and levels of knowledge
- To familiarize with the critical elements of KM



Concept of Knowledge

The Concept of Knowledge

Please Highlight the Picture for Details

Timeline: Antiquity 4000 - 2400BC

Definition of knowledge: ?

Knowledge model: Narratives and symbolic depictions

Important discovery: No written record



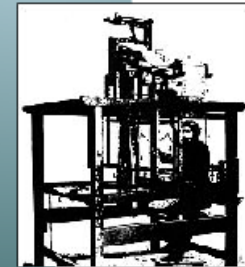
Antiquity
4000-2400BC



Greek
Philosophers
800BC-500AD



Medieval/
Middle Age
500AD-1400s



Industrial
revolution
1500s-1800s



Modern Era
1900 onwards

Concept of Knowledge

The Concept of Knowledge

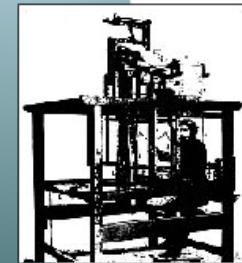
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Timeline: Greek Philosophers 800BC - 500AD

Definition of Knowledge: Knowledge is justified true belief

Knowledge model: Mathematics and logic

Important discovery: Geometry, astronomy, and Aristotelian logical evidence



Concept of Knowledge

The Concept of Knowledge

Please Highlight the Picture for Details

Timeline: Medieval / Middle Age 500AD - 1400s

Definition of Knowledge: Knowledge is profess and apprentice

Knowledge model: Scholastic method

Important discovery: Practical manuals and formulation of knowledge that can still be found at present



Antiquity
4000-2400BC



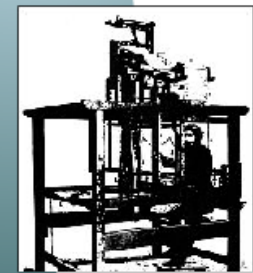
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1500s-1800s



Concept of Knowledge

The Concept of Knowledge

Please Highlight the Picture for Details

Timeline: Industrial revolution 1500s - 1800s

Definition of Knowledge: Knowledge is power empowering effective action

Knowledge model: Scientific method

Important discovery: Testing knowledge by proven statistical means for validation



Antiquity
4000-2400BC



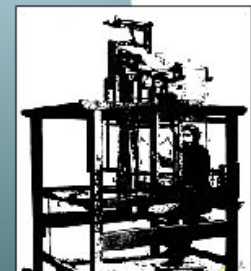
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Concept of Knowledge

The Concept of Knowledge

Please Highlight the Picture for Details

Timeline: Modern Era 1900 onwards

Definition of Knowledge: A part of the market-economy system

Knowledge model: Learning organization

Important discovery: Knowledge arises through interpretation or experience, abstraction, and conceptualisation



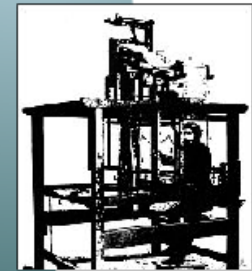
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Data, Information & Knowledge

Data

- Data is a set of discrete, objective facts about events

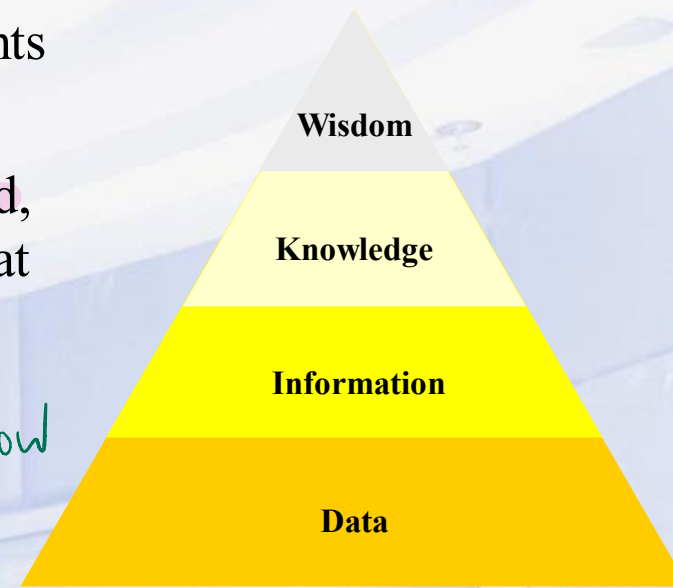
Information

- Processed (contextualised, categorised, **calculated**, corrected, and condensed) data to show & display what is going on

Knowledge

5W1H What, why, Where, Who, When, How

- Organised/synthesised information to explain or predict events & new situations, to make connections between what is happening now and what happens next, and to make decisions
- Knowledge is often a mix of contextual information, framed experience, expert insight values & beliefs that provides a framework for evaluating & incorporating new experiences & information



The Knowledge Pyramid



~~*~~ In-class Exercise 1: Case Study

The outbreak of the Severe Acute Respiratory Syndrome (SARS) has caused much public concern. The issues have been reported through various channels such as TV, radio, internet, newspapers, etc. Hong Kong has introduced a rigorous contact tracing procedure and good practice. All close contacts of known SARS cases are quarantined at home. In addition, their Hong Kong Identity Card numbers are passed to the Immigration Department to ensure that such people cannot leave the territory. Contact tracing has established epidemiological links for the majority of Hong Kong's SARS patients. Around 8.6% of patients have no identifiable exposure source. Detailed investigation is ongoing.

As of today, a total increase of 168 new cases and 26 deaths have been reported. The distribution of the new deaths were reported in China (17 cases), HKSAR (6 cases), Taiwan, China (2 Cases) and Singapore (1 Case).” *distribution*

(Source: http://www.who.int/csr/don/2003_05_12/en)

纯数字



Discussion question

Based on the above scenario, identify the difference between data, information, and knowledge. Give an example for each ONE of them.

↓ whole statement



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Explicit vs. Tacit Knowledge

Explicit knowledge

- Easier to identify
- Re-usable in consistent and repeatable manner - for decision-making and/or exercising judgement
- Can be stored as a written procedure or as a process in a computer system
- Stored as **artefacts** - artificial physical or virtual entity that can be measured, identified, distributed and audited

Tacit knowledge

- Lessons learned, methodologies, cases, stories
- Personal, context-specific, hard to formalise and communicate
- Intuitions, rules of thumb, mind-sets, unwritten rules of turf and territory, unconscious values
- Trivial fundamental philosophy